

# AI WeatherWise

## AI-Augmented Backend Development Project

### PHASE 2 – REQUIREMENT ANALYSIS

|               |                                    |
|---------------|------------------------------------|
| Team No.      | 05                                 |
| Team ID       | SWTID-2026-9139                    |
| Project Title | AI WeatherWise                     |
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#### 1. Requirement Analysis

The Requirement Analysis phase identified the functional and technical requirements needed to develop AI WeatherWise as a secure backend application. The team studied user needs, system behaviour, data flow, and external service requirements before proceeding to the design phase.

##### Project Goal

Develop a RESTful backend application that securely manages users, retrieves weather information, stores favourite locations, and provides AI-powered weather recommendations.

#### 2. User Requirements

The system was analysed from the user's perspective to ensure that every feature solves a practical requirement.

| Requirement         | Purpose                           |
|---------------------|-----------------------------------|
| User Registration   | Create a secure account           |
| User Login          | Authenticate using JWT            |
| Current Weather     | View live weather information     |
| Favourite Locations | Save multiple cities              |
| Recommendations     | Receive weather-based suggestions |

### 3. Functional Requirements

- User registration with encrypted password storage.
- JWT-based login authentication.
- Weather data retrieval using external APIs.
- Save, update, and manage favourite locations.
- Generate AI-powered recommendations.
- Centralized error handling and validation.

### 4. Non-Functional Requirements

#### Security

JWT authentication, hashed passwords, protected routes, and input validation ensure secure backend operations.

#### Performance

The backend should respond quickly to API requests while maintaining efficient database operations.

#### Scalability

The modular architecture allows future expansion without major code restructuring.

## 5. System Requirements

| Component      | Technology               |
|----------------|--------------------------|
| Runtime        | Node.js                  |
| Framework      | Express.js               |
| Database       | MongoDB & Mongoose       |
| Authentication | JWT                      |
| Testing        | Thunder Client / Postman |

## 6. Constraints and Assumptions

The project depends on external Weather APIs and AI services. Internet connectivity is required for live weather retrieval. API limitations may affect responses, so fallback handling is included in the backend.

## 7. Requirement Analysis Outcome

At the end of this phase, the team finalized the complete list of functional requirements, non-functional requirements, user needs, technical requirements, and system constraints. These requirements became the foundation for the Project Design phase.